Spring Boot – Sample Learning Document

A text-based sample PDF for testing PDF extraction in a LangChain RAG application.

1. What is Spring Boot?

Spring Boot is a Java framework built on the Spring ecosystem. It simplifies application development through auto-configuration, starter dependencies, and embedded server support. Developers can create standalone applications with less manual configuration.

2. Important Features

Auto-configuration: Configures common components based on dependencies and application settings.

Starter dependencies: Group libraries commonly used for a particular task. spring-boot-starter-web provides dependencies for web applications.

Embedded server: Applications can run with an embedded server such as Tomcat, without requiring a separately managed application server.

Externalized configuration: Settings can be supplied through application.properties, environment variables, or command-line arguments.

3. Creating a REST API

A REST controller handles HTTP requests and returns data to clients. Here is a simple GET endpoint:

```
@RestController
@RequestMapping("/api")
public class HelloController {

@GetMapping("/hello")
public String hello() {
return "Hello from Spring Boot";
}
}
```

When the application runs on port 8080, a client can send a GET request to http://localhost:8080/api/hello.

4. Dependency Injection

Dependency injection is a core Spring concept. A class declares the dependencies it needs, and the Spring container provides them. Constructor injection makes required dependencies explicit and supports unit testing.

5. application.properties

```
server.port=8081
spring.application.name=sample-service
```

The server.port property changes the embedded server port from 8080 to 8081. The application name can be useful in logs and service discovery.

6. Spring Boot and Microservices

A microservice is an independently deployable application focused on a business capability. Spring Boot is commonly used to build microservices because it provides web support, configuration, dependency injection, and production integration options.

Services may communicate synchronously through HTTP REST APIs or asynchronously through messaging systems such as Apache Kafka. Services should handle network failures, timeouts, and retries carefully.

7. Database Access

Spring Data JPA simplifies database access through repository abstractions. A repository can provide operations such as save, findById, findAll, and deleteById. Hibernate is a commonly used JPA implementation.

8. Quick Revision

• Spring Boot reduces setup and configuration effort.
• Auto-configuration uses dependencies and configuration properties.
• REST controllers commonly use @RestController and @GetMapping.
• Constructor injection provides required dependencies.
• Spring Boot supports embedded servers and independently deployable services.